Number Patterns

Name _____ **Date** _____ No calculator Version 1

KEY WORDS pattern · term · position · constant change · rule · square number · cube number · triangular number

A Continuing a pattern ●○○ — ANSWER ON THE SHEET

Remember: In a linear pattern the same amount is added or subtracted each time.

1. Write the next two terms of each pattern.

a) 3, 7, 11, 15, ___, ___

b) 5, 11, 17, 23, ___, ___

c) 20, 17, 14, 11, ___, ___

d) 1, 3.5, 6, 8.5, ___, ___

e) −10, −6, −2, 2, ___, ___

f) 50, 43, 36, 29, ___, ___

g) 0.2, 0.5, 0.8, 1.1, ___, ___

h) 8, 19, 30, 41, ___, ___

B The constant change ●○○ — WORK IN YOUR BOOK

2. What is added or subtracted each time?

a) 4, 7, 10, 13, 16

b) 10, 18, 26, 34, 42

c) 30, 25, 20, 15, 10

d) 7, 7.5, 8, 8.5, 9

e) −6, −4, −2, 0, 2

f) 45, 36, 27, 18, 9

C Decreasing patterns ●●○ — WORK IN YOUR BOOK

Watch out: A decreasing pattern can go below zero. After 2 comes -1 if you are subtracting 3 each time.

3. Each pattern keeps decreasing. Find the first term that is less than zero.

a) 10, 7, 4, 1, ...

b) 18, 13, 8, 3, ...

c) 7, 5, 3, 1, ...

d) 30, 22, 14, 6, ...

D Using a rule ●●○ — WORK IN YOUR BOOK

Remember: In the rule $T = 3n + 2$, n is the position of the term. For the 4th term, put $n = 4$: $3 \times 4 + 2 = 14$.

4. Write the first four terms of the pattern with each rule.

a) $T = 3n + 2$

b) $T = 5n - 1$

c) $T = 2n + 7$

d) $T = 4n$

e) $T = 10n - 6$

f) $T = -2n + 20$

5. Find the term asked for.

a) $T = 3n + 2$, the 10th term

b) $T = 5n - 1$, the 20th term

c) $T = 4n + 7$, the 50th term

d) $T = -2n + 100$, the 30th term

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E Finding the rule ●●● — WORK IN YOUR BOOK

> **Remember:** The number in front of n is the constant change. Then work out what must be added or subtracted to make the first term correct.

6. Find the rule for the nth term of each pattern.

a) 5, 8, 11, 14, 17

b) 7, 9, 11, 13, 15

c) 4, 8, 12, 16, 20

d) 1, 6, 11, 16, 21

e) 10, 16, 22, 28, 34

f) 20, 18, 16, 14, 12

F Patterns in context ●●● — WORK IN YOUR BOOK

7. Squares are made in a row from matchsticks. The table shows the number of matchsticks in each pattern.

Pattern	1	2	3	4
Matchsticks	4	7	10	13

a) Find the number for pattern 5.

b) Write the rule for pattern n.

c) Find the number for pattern 20.

d) Which pattern has 46 matchsticks?

G Square, cube and triangular numbers ●●○ — ANSWER ON THE SHEET

> **Remember:** Square numbers: $1^2, 2^2, 3^2, \ldots$ Cube numbers: $1^3, 2^3, 3^3, \ldots$ Triangular numbers: $1,\ 1+2,\ 1+2+3,\ \ldots$

8. Write each list of numbers.

a) The first 8 square numbers _____

b) The first 5 cube numbers _____

c) The first 7 triangular numbers _____

9. Write the next two numbers in each pattern.

a) 1, 4, 9, 16, ___, ___

b) 1, 3, 6, 10, ___, ___

c) 1, 8, 27, ___, ___

10. Is each number square, cube or triangular? Some are more than one, and some are none of them.

a) 15 _____ **b)** 49 _____ **c)** 64 _____ **d)** 21 _____

e) 36 _____ **f)** 125 _____ **g)** 50 _____ **h)** 1 _____

★ Extension — FOR EARLY FINISHERS

11. Find the 10th triangular number.

12. Add the triangular numbers 6 and 10, which are next to each other in the list. What kind of number do you get? Try another pair.

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Number Patterns

Name _____ **Date** _____ No calculator Version 2

KEY WORDS pattern · term · position · constant change · rule · square number · cube number · triangular number

A Continuing a pattern ●○○ — ANSWER ON THE SHEET

Remember: In a linear pattern the same amount is added or subtracted each time.

1. Write the next two terms of each pattern.

a) 2, 7, 12, 17, ___, ___

b) 4, 11, 18, 25, ___, ___

c) 25, 21, 17, 13, ___, ___

d) 2, 3.5, 5, 6.5, ___, ___

e) −12, −7, −2, 3, ___, ___

f) 60, 51, 42, 33, ___, ___

g) 0.1, 0.5, 0.9, 1.3, ___, ___

h) 7, 19, 31, 43, ___, ___

B The constant change ●○○ — WORK IN YOUR BOOK

2. What is added or subtracted each time?

a) 5, 9, 13, 17, 21

b) 12, 21, 30, 39, 48

c) 40, 34, 28, 22, 16

d) 3, 3.5, 4, 4.5, 5

e) −8, −5, −2, 1, 4

f) 54, 46, 38, 30, 22

C Decreasing patterns ●●○ — WORK IN YOUR BOOK

Watch out: A decreasing pattern can go below zero. After 2 comes -1 if you are subtracting 3 each time.

3. Each pattern keeps decreasing. Find the first term that is less than zero.

a) 11, 8, 5, 2, ...

b) 17, 12, 7, 2, ...

c) 9, 7, 5, 3, ...

d) 28, 20, 12, 4, ...

D Using a rule ●●○ — WORK IN YOUR BOOK

Remember: In the rule $T = 3n + 2$, n is the position of the term. For the 4th term, put $n = 4$: $3 \times 4 + 2 = 14$.

4. Write the first four terms of the pattern with each rule.

a) $T = 4n + 1$

b) $T = 3n - 2$

c) $T = 2n + 9$

d) $T = 6n$

e) $T = 10n - 4$

f) $T = -3n + 30$

5. Find the term asked for.

a) $T = 4n + 1$, the 10th term

b) $T = 3n - 2$, the 20th term

c) $T = 6n + 5$, the 50th term

d) $T = -3n + 200$, the 40th term

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E Finding the rule ●●● — WORK IN YOUR BOOK

> **Remember:** The number in front of n is the constant change. Then work out what must be added or subtracted to make the first term correct.

6. Find the rule for the nth term of each pattern.

a) 6, 10, 14, 18, 22

b) 5, 7, 9, 11, 13

c) 3, 6, 9, 12, 15

d) 2, 7, 12, 17, 22

e) 9, 16, 23, 30, 37

f) 30, 27, 24, 21, 18

F Patterns in context ●●● — WORK IN YOUR BOOK

7. Triangles are made in a row from matchsticks. The table shows the number of matchsticks in each pattern.

Pattern	1	2	3	4
Matchsticks	3	5	7	9

a) Find the number for pattern 5.

b) Write the rule for pattern n.

c) Find the number for pattern 25.

d) Which pattern has 37 matchsticks?

G Square, cube and triangular numbers ●●○ — ANSWER ON THE SHEET

> **Remember:** Square numbers: $1^2, 2^2, 3^2, \ldots$ Cube numbers: $1^3, 2^3, 3^3, \ldots$ Triangular numbers: $1,\ 1+2,\ 1+2+3,\ \ldots$

8. Write each list of numbers.

a) The first 9 square numbers _____

b) The first 5 cube numbers _____

c) The first 8 triangular numbers _____

9. Write the next two numbers in each pattern.

a) 1, 4, 9, 16, 25, ___, ___ **b)** 1, 3, 6, 10, 15, ___, ___ **c)** 1, 8, ___, ___

10. Is each number square, cube or triangular? Some are more than one, and some are none of them.

a) 10 _____ **b)** 81 _____ **c)** 64 _____ **d)** 28 _____

e) 36 _____ **f)** 27 _____ **g)** 40 _____ **h)** 1 _____

★ Extension — FOR EARLY FINISHERS

11. Find the 12th triangular number.

12. Add the triangular numbers 10 and 15, which are next to each other in the list. What kind of number do you get? Try another pair.

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